

Enterprise Healthcare Content Intelligence Using Retrieval-Augmented Generation (RAG) and Local Large Language Models for Policy-to-Code Automation

Executive Summary

Healthcare organizations generate and manage enormous volumes of unstructured documents, including billing and coding policies, clinical practice guidelines, payer-provider contracts, reimbursement manuals, regulatory publications, and healthcare compliance documentation. These documents are frequently updated and contain complex operational rules that directly influence reimbursement accuracy, utilization management, coding compliance, claims processing, and clinical decision-making. Manual interpretation of these documents is labor-intensive, time-consuming, expensive, and often results in inconsistent policy interpretation.

Recent advances in Artificial Intelligence (AI), particularly Retrieval-Augmented Generation (RAG), semantic search, document intelligence, and local Large Language Models (LLMs), provide new opportunities to automate healthcare knowledge management while maintaining privacy, transparency, and regulatory compliance. Rather than relying solely on model memory, RAG enables AI systems to retrieve relevant evidence directly from trusted documents before generating responses, thereby improving explainability and reducing hallucinations.

To demonstrate the practical feasibility of these technologies, a lightweight Proof of Concept (PoC) named the Enterprise Health Intelligence Engine was developed. The system ingests healthcare documents, semantically indexes their contents, retrieves relevant passages, summarizes policies into structured schemas, compares revisions between policy versions, converts narrative healthcare policies into executable Python business rules, and validates generated rules using representative patient scenarios. All AI inference is performed locally using Ollama-hosted Llama 3.1 models, ensuring sensitive healthcare information remains entirely within the organization's infrastructure.

Current Trends in Healthcare Content Intelligence

Healthcare organizations are increasingly transitioning from traditional keyword-based document search toward intelligent document understanding powered by semantic retrieval. Studies estimate that nearly 80% of healthcare information exists as unstructured text, including contracts, policy manuals, scanned documents, and clinical guidelines, making efficient information retrieval a growing challenge.

Organizations are also adopting local AI deployment models to minimize privacy risks associated with external cloud services and to satisfy stringent regulatory requirements governing protected health information (PHI). Local deployment provides greater organizational control over sensitive healthcare data while enabling AI-assisted knowledge management.

Another major trend is the movement toward structured knowledge extraction. Instead of storing policy information solely as narrative text, organizations increasingly transform healthcare policies into structured, machine-readable formats that can support automated claims processing, decision support systems, workflow orchestration, and payment integrity operations.

Simultaneously, payer policies continue to evolve rapidly. Filing deadlines, reimbursement rules, prior authorization requirements, coding penalties, and contractual obligations are frequently revised. These continual changes create significant operational challenges, requiring healthcare organizations to efficiently identify policy revisions and translate updated business rules into production systems without introducing costly errors.

Opportunities and Strategic Value

The proposed architecture offers numerous strategic opportunities for healthcare analytics

organizations such as Cotiviti.

AI-powered semantic search enables faster analysis of clinical guidelines, reimbursement policies, and payer-provider contracts while improving consistency in policy interpretation. Automated policy summarization reduces manual review effort and accelerates onboarding of coding specialists, auditors, and compliance personnel.

The system supports automatic comparison of policy revisions using semantic change detection, enabling organizations to rapidly identify modifications such as shortened filing deadlines, altered reimbursement criteria, new prior authorization requirements, and revised financial penalties. Such contract drift analytics can proactively identify increased financial liability and operational risk. A particularly valuable capability is Policy-to-Code Automation, where narrative healthcare policies are automatically transformed into structured JSON representations and subsequently compiled into executable business logic. This deterministic two-stage workflow significantly reduces the time required to update payment integrity systems and claims processing engines.

Structured information extraction is enforced through Pydantic schema validation, ensuring that unstructured policy documents are consistently converted into standardized objects containing policy titles, billing codes, prerequisites, reimbursement rules, penalties, and other operational parameters with uniform template alignment.

By executing all AI models locally using open-source LLMs, organizations can preserve data privacy, eliminate dependence on external APIs, and maintain full control over confidential healthcare documentation.

Risk Management

Although AI substantially improves operational efficiency, several technical and governance challenges remain.

Large Language Models may generate hallucinated information if retrieval grounding is insufficient. This risk is mitigated using strictly bounded RAG prompting, which constrains model responses exclusively to retrieved document evidence stored in the local vector database.

Automatically generated business rules require expert human validation before deployment, particularly in highly regulated healthcare payment environments where incorrect recommendations may result in financial loss or compliance violations.

Since healthcare policies are frequently revised, continuous document ingestion, semantic re-indexing, and version management are necessary to ensure retrieval accuracy. Regulatory compliance also demands comprehensive governance over AI-generated recommendations.

Another significant consideration is runtime security. Because Policy-to-Code conversion generates executable Python logic, the resulting code must execute exclusively inside isolated namespace sandbox environments to prevent code injection attacks or unintended access to production systems.

Strategic Recommendations for Cotiviti

Cotiviti could leverage this architecture to build next-generation AI-powered healthcare content intelligence platforms that improve operational efficiency and payment accuracy.

Recommended strategic initiatives include developing enterprise-scale policy intelligence platforms supporting payer and provider documentation, automated monitoring of policy revisions with semantic change detection, and structured rule extraction pipelines capable of converting regulatory documents into executable business logic.

Cotiviti could also commercialize a dual-stage Policy-to-Code synthesis engine that first extracts structured parameter maps and subsequently compiles them into deterministic business rules, dramatically reducing the lifecycle for updating payment accuracy systems.

Another promising opportunity is the development of a Contract Drift Analytics Suite capable of continuously monitoring year-over-year payer contract amendments and automatically highlighting changes in reimbursement policies, coding requirements, filing deadlines, prior authorization rules,

and financial penalties.

Finally, adopting standardized local open-source language models deployed through containerized on-premises infrastructure would strengthen data privacy, reduce external data leakage risks, and provide scalable AI capabilities aligned with healthcare compliance requirements.

Overall, the demonstrated Proof of Concept illustrates how Retrieval-Augmented Generation, semantic search, local Large Language Models, and structured document intelligence can substantially reduce manual effort while improving consistency, transparency, operational efficiency, and regulatory compliance. Although expert oversight remains essential, these technologies have strong potential to become foundational decision-support tools for healthcare analytics organizations such as Cotiviti.

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